

A Project Report On Employee Management System

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CERTIFICATE

This is to certify that the project entitled “Employee Management system” is a bonafide report of the work carried out by

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of Department of Information Technology, semester IV, under the guidance and supervision for the subject Database Management System. They were involved in Project training during the academic year 2024-2025.

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1.SYSTEM OVERVIEW

Data is the code word of the computer industry. Data refers to a collection of facts usually collected as a result of observation and experiment or processes within a computer system. This may consist of numbers, words or images or observations of a set of variables. Data are often viewed as a lowest level of abstraction from which information and knowledge are derived.

The **Employee Management System (EMS)** is **database driven system** that designed to manage employee data efficiently within organization. It automates various HR functions such as hiring, payroll, attendance tracking and performance evaluation. The system ensures smooth operations by providing a centralized platform for storing and managing employee-related data securely.

1.3 ADVANTAGES OF THE PROPOSED SYSTEM

1.Improved Data Management:

- i) Centralized Database – Stores all employee-related data in one place, making retrieval and updates easier.
- ii) Efficient Record Keeping – Eliminates the need for manual paperwork, reducing errors and redundancies.

2. Automation & Efficiency

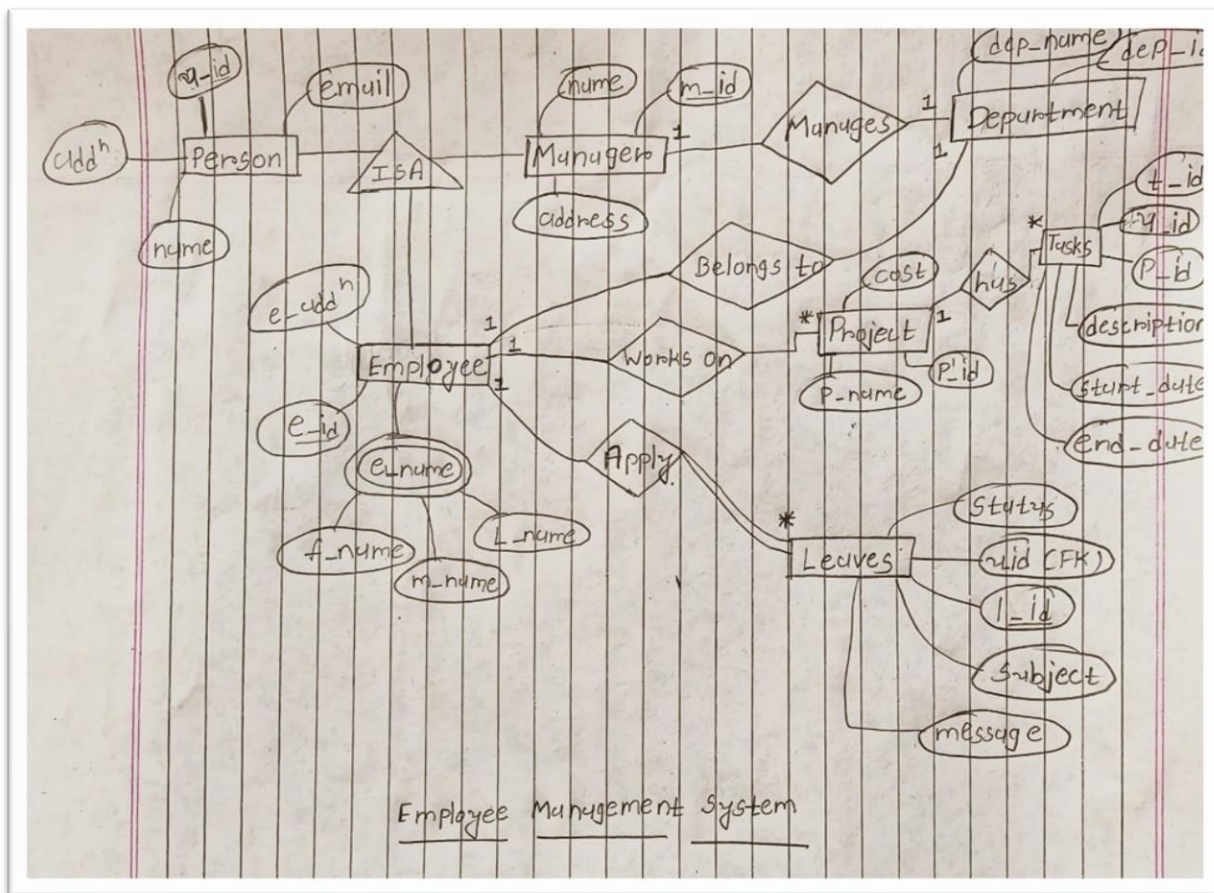
- i) Reduces Manual Work – Automates HR processes like attendance and leave management.
- ii) Faster Data Processing – Speeds up employee record retrieval, payroll processing, and performance tracking.

3. Enhanced Security

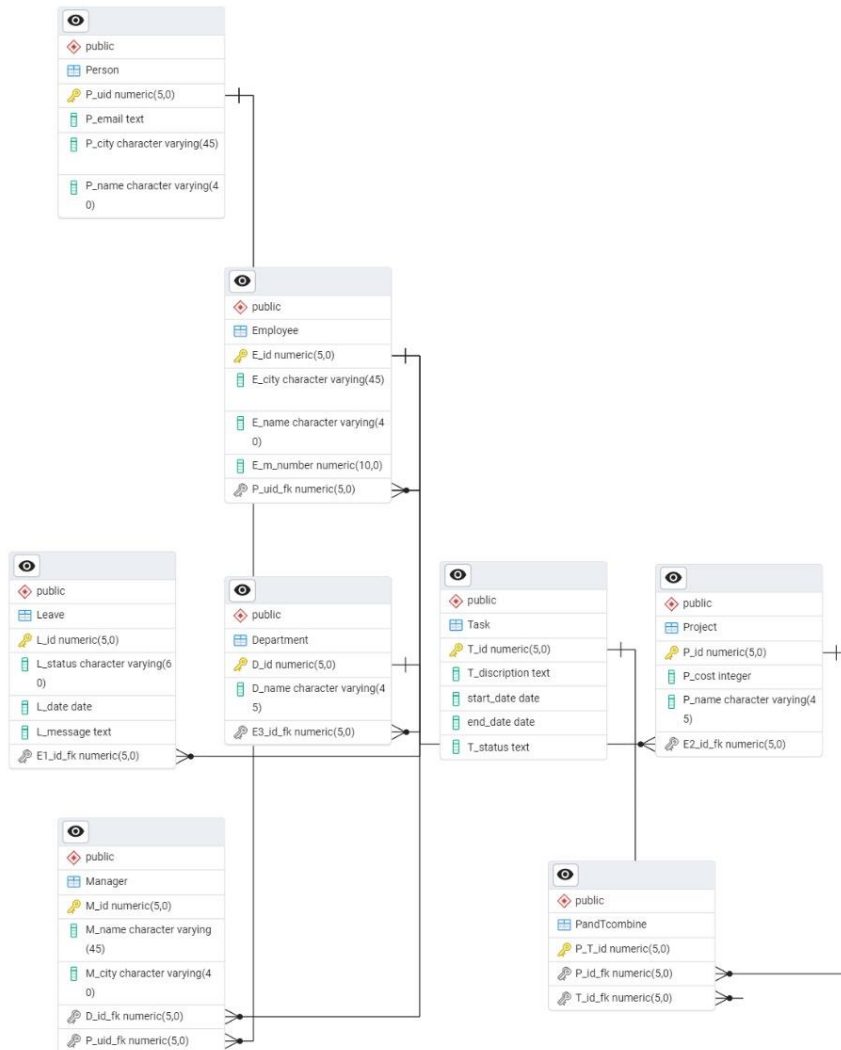
- i) Role-Based Access Control (RBAC) – Ensures only authorized users can access or modify specific data.

ii) Data Encryption – Protects sensitive information such as salaries, personal details, and performance records.

2. ENTITY-RELATIONSHIP MODEL



3. RELATIONAL SCHEMA



4. DATA DICTIONARY

4.1 PERSON

```
Employee_Manager_Management_System=# \d "Person"
Table "public.Person"
Column |          Type          | Collation | Nullable | Default
-----|-----|-----|-----|-----
P_uid  | numeric(5,0)           |           | not null |
P_email | text                   |           |          |
P_city | character varying(45)  |           |          |
P_name | character varying(40)  |           |          |
Indexes:
    "Person_pkey" PRIMARY KEY, btree ("P_uid")
Referenced by:
    TABLE "Manager" CONSTRAINT "P_uid_fk" FOREIGN KEY ("P_uid_fk") REFERENCES "Person"("P_uid")
    TABLE "Employee" CONSTRAINT "P_uid_fk" FOREIGN KEY ("P_uid_fk") REFERENCES "Person"("P_uid") NOT VALID
```

4.2 EMPLOYEE

```
Employee_Manager_Management_System=# \d "Employee"
Table "public.Employee"
Column |          Type          | Collation | Nullable | Default
-----|-----|-----|-----|-----
E_id   | numeric(5,0)           |           | not null |
E_city | character varying(45)  |           | not null |
E_name | character varying(40)  |           | not null |
E_m_number | numeric(10,0)        |           | not null |
P_uid_fk | numeric(5,0)           |           | not null |
Indexes:
    "Employee_pkey" PRIMARY KEY, btree ("E_id")
Foreign-key constraints:
    "P_uid_fk" FOREIGN KEY ("P_uid_fk") REFERENCES "Person"("P_uid") NOT VALID
Referenced by:
    TABLE "Leave" CONSTRAINT "E1_id_fk" FOREIGN KEY ("E1_id_fk") REFERENCES "Employee"("E_id")
    TABLE "Project" CONSTRAINT "E2_id_fk" FOREIGN KEY ("E2_id_fk") REFERENCES "Employee"("E_id")
    TABLE "Department" CONSTRAINT "E3_id_fk" FOREIGN KEY ("E3_id_fk") REFERENCES "Employee"("E_id")
```

4.3 MANAGER

```
Employee_Manager_Management_System=# \d "Manager"
Table "public.Manager"
Column |          Type          | Collation | Nullable | Default
-----|-----|-----|-----|-----
M_id   | numeric(5,0)           |           | not null |
M_name | character varying(45)  |           | not null |
M_city | character varying(40)  |           | not null |
D_id_fk | numeric(5,0)           |           | not null |
P_uid_fk | numeric(5,0)           |           | not null |
Indexes:
    "M_id" PRIMARY KEY, btree ("M_id")
Foreign-key constraints:
    "D_id_fk" FOREIGN KEY ("D_id_fk") REFERENCES "Department"("D_id")
    "P_uid_fk" FOREIGN KEY ("P_uid_fk") REFERENCES "Person"("P_uid")
```

4.4 DEPARTMENT

```
Employee_Manager_Management_System=# \d "Department"
Table "public.Department"
  Column |          Type          | Collation | Nullable | Default
-----+-----+-----+-----+-----
D_id    | numeric(5,0)           |           | not null |
D_name  | character varying(45)  |           | not null |
E3_id_fk| numeric(5,0)           |           | not null |
Indexes:
    "D_id" PRIMARY KEY, btree ("D_id")
Foreign-key constraints:
    "E3_id_fk" FOREIGN KEY ("E3_id_fk") REFERENCES "Employee"("E_id")
Referenced by:
    TABLE "Manager" CONSTRAINT "D_id_fk" FOREIGN KEY ("D_id_fk") REFERENCES "Department"("D_id")
```

4.5 PROJECT

```
Employee_Manager_Management_System=# \d "Project"
Table "public.Project"
  Column |          Type          | Collation | Nullable | Default
-----+-----+-----+-----+-----
P_id    | numeric(5,0)           |           | not null |
P_cost  | integer                |           | not null |
P_name  | character varying(45)  |           | not null |
E2_id_fk| numeric(5,0)           |           | not null |
Indexes:
    "P_id" PRIMARY KEY, btree ("P_id")
Foreign-key constraints:
    "E2_id_fk" FOREIGN KEY ("E2_id_fk") REFERENCES "Employee"("E_id")
Referenced by:
    TABLE "PandTcombine" CONSTRAINT "P_id_fk" FOREIGN KEY ("P_id_fk") REFERENCES "Project"("P_id")
```

4.6 TASKS

```
Employee_Manager_Management_System=# \d "Task"
Table "public.Task"
  Column |          Type          | Collation | Nullable | Default
-----+-----+-----+-----+-----
T_id    | numeric(5,0)           |           | not null |
T_discription | text                |           | not null |
start_date | date                |           | not null |
end_date  | date                |           | not null |
T_status  | text                |           | not null |
p_id_fk  | numeric(5,0)         |           |          |
Indexes:
    "T_id" PRIMARY KEY, btree ("T_id")
Foreign-key constraints:
    "p_id_fk" FOREIGN KEY (p_id_fk) REFERENCES "Project"("P_id") NOT VALID
Referenced by:
    TABLE "PandTcombine" CONSTRAINT "T_id_fk" FOREIGN KEY ("T_id_fk") REFERENCES "Task"("T_id")
```

4.7 LEAVES

Employee_Manager_Management_System=# \d "Leave"

Table "public.Leave"

Column	Type	Collation	Nullable	Default
L_id	numeric(5,0)		not null	
L_status	character varying(60)		not null	
L_date	date		not null	
L_message	text		not null	
E1_id_fk	numeric(5,0)		not null	

Indexes:

"L_id" PRIMARY KEY, btree ("L_id")

Foreign-key constraints:

"E1_id_fk" FOREIGN KEY ("E1_id_fk") REFERENCES "Employee"("E_id")

4.8 TASKS AND PROJECT COMBINE

Employee_Manager_Management_System=# \d "PandTcombine"

Table "public.PandTcombine"

Column	Type	Collation	Nullable	Default
P_T_id	numeric(5,0)		not null	
P_id_fk	numeric(5,0)		not null	
T_id_fk	numeric(5,0)		not null	

Indexes:

"P_T_id" PRIMARY KEY, btree ("P_T_id")

Foreign-key constraints:

"P_id_fk" FOREIGN KEY ("P_id_fk") REFERENCES "Project"("P_id")

"T_id" FOREIGN KEY ("T_id_fk") REFERENCES "Task"("T_id")

5. DATA IMPLEMENTATION

A) SCHEMA

5.1.1 PERSON

CREATE TABLE Person (

"P_uid" numeric(5,0) NOT NULL,

"P_email" text COLLATE pg_catalog."default",

"P_city" character varying(45) COLLATE pg_catalog."default",

"P_name" character varying(40) COLLATE pg_catalog."default",

CONSTRAINT "Person_pkey" PRIMARY KEY ("P_uid")

);

5.1.2 Employee

```

CREATE TABLE Employee (
    "E_id" numeric(5,0) NOT NULL,
    "E_city" character varying(45) COLLATE pg_catalog."default" NOT NULL,
    "E_name" character varying(40) COLLATE pg_catalog."default" NOT NULL,
    "E_m_number" numeric(10,0) NOT NULL,
    "P_uid_fk" numeric(5,0) NOT NULL,
    CONSTRAINT "Employee_pkey" PRIMARY KEY ("E_id"),
    CONSTRAINT "P_uid_fk" FOREIGN KEY ("P_uid_fk")
        REFERENCES public."Person" ("P_uid") MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION
        NOT VALID
);

```

5.1.3 Manager

```

CREATE TABLE Manager (
    "M_id" numeric(5,0) NOT NULL,
    "M_name" character varying(45) COLLATE pg_catalog."default" NOT NULL,
    "M_city" character varying(40) COLLATE pg_catalog."default" NOT NULL,
    "D_id_fk" numeric(5,0) NOT NULL,
    "P_uid_fk" numeric(5,0) NOT NULL,
    CONSTRAINT "M_id" PRIMARY KEY ("M_id"),
    CONSTRAINT "D_id_fk" FOREIGN KEY ("D_id_fk")
        REFERENCES public."Department" ("D_id") MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION,
    CONSTRAINT "P_uid_fk" FOREIGN KEY ("P_uid_fk")
        REFERENCES public."Person" ("P_uid") MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION
);

```

);

5.1.4 Department

```
CREATE TABLE Department(  
    "D_id" numeric(5,0) NOT NULL,  
    "D_name" character varying(45) COLLATE pg_catalog."default" NOT NULL,  
    "E3_id_fk" numeric(5,0) NOT NULL,  
    CONSTRAINT "D_id" PRIMARY KEY ("D_id"),  
    CONSTRAINT "E3_id_fk" FOREIGN KEY ("E3_id_fk")  
        REFERENCES public."Employee" ("E_id") MATCH SIMPLE  
        ON UPDATE NO ACTION  
        ON DELETE NO ACTION  
);
```

5.1.5 Project

```
CREATE TABLE Project (  
    "P_id" numeric(5,0) NOT NULL,  
    "P_cost" integer NOT NULL,  
    "P_name" character varying(45) COLLATE pg_catalog."default" NOT NULL,  
    "E2_id_fk" numeric(5,0) NOT NULL,  
    CONSTRAINT "P_id" PRIMARY KEY ("P_id"),  
    CONSTRAINT "E2_id_fk" FOREIGN KEY ("E2_id_fk")  
        REFERENCES public."Employee" ("E_id") MATCH SIMPLE  
        ON UPDATE NO ACTION  
        ON DELETE NO ACTION  
);
```

5.1.6 TASKS

```
CREATE TABLE Task(  
    "T_id" numeric(5,0) NOT NULL,  
    "T_discription" text COLLATE pg_catalog."default" NOT NULL,
```

```

start_date date NOT NULL,
end_date date NOT NULL,
"T_status" text COLLATE pg_catalog."default" NOT NULL,
p_id_fk numeric(5,0),
CONSTRAINT "T_id" PRIMARY KEY ("T_id"),
CONSTRAINT p_id_fk FOREIGN KEY (p_id_fk)
    REFERENCES public."Project" ("P_id") MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE NO ACTION
    NOT VALID
);

```

5.1.7 LEAVES

```

CREATE TABLE Leave(
    "L_id" numeric(5,0) NOT NULL,
    "L_status" character varying(60) COLLATE pg_catalog."default" NOT NULL,
    "L_date" date NOT NULL,
    "L_message" text COLLATE pg_catalog."default" NOT NULL,
    "E1_id_fk" numeric(5,0) NOT NULL,
    CONSTRAINT "L_id" PRIMARY KEY ("L_id"),
    CONSTRAINT "E1_id_fk" FOREIGN KEY ("E1_id_fk")
        REFERENCES public."Employee" ("E_id") MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION
);

```

5.1.8 TASK AND PROJECT COMBINE

```

CREATE TABLE PandTcombine (
    "P_T_id" numeric(5,0) NOT NULL,
    "P_id_fk" numeric(5,0) NOT NULL,

```

```

"T_id_fk" numeric(5,0) NOT NULL,
CONSTRAINT "P_T_id" PRIMARY KEY ("P_T_id"),
CONSTRAINT "P_id_fk" FOREIGN KEY ("P_id_fk")
    REFERENCES public."Project" ("P_id") MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE NO ACTION,
CONSTRAINT "T_id" FOREIGN KEY ("T_id_fk")
    REFERENCES public."Task" ("T_id") MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE NO ACTION
);

```

B) DATA INSERTION

5.2.1 Person

```

INSERT INTO Person (
    "P_uid", "P_email", "P_city", "P_name")
VALUES ('108','raj1@gmail.com','Surat','Raj Shah'),
('109','yaj1@gmail.com','Rajkot','Jay Sakariya'),
('110','yug3@gmail.com','Surendranagar','Yug Sakariya'),
('107','jaypal4@gmail.com','Nadiad','Jaypal Soni'),
('106','sandip5@gmail.com','Valsad','Sandip Sojitra'),
('105','rahul3@gmail.com','Jamnagar','Rahul Sakariya'),
('104','dev5@gmail.com','navsari','Dev Ansari'),
('103','deven1@gmail.com','Surat','Deven Gol'),
('102','om12@gmail.com','Ahmedabad','Om Jani'),
('101','jatin7@gmail.com','Palitana','Jatin Mandali');

```

5.2.2 Employee

```
INSERT INTO Employee (  
"E_id", "E_city", "E_name", "E_m_number", "P_uid_fk")  
VALUES ('201', 'Ahmedabad', 'Akshat Jani', '9874567813', '101'),  
( '202', 'Surat', 'Yusuf Pathan', '9745786165', '102'),  
( '203', 'Nadiad', 'Irfan Pathan', '9587456325', '101'),  
( '204', 'Vadodara', 'Suresh Raina', '9987563214', '103'),  
( '205', 'Rajkot', 'Vinod Kambali', '9875463215', '103'),  
( '206', 'Bhuj', 'Vedant Patel', '9321456854', '106'),  
( '207', 'Dang', 'Ansh Thummar', '9658742145', '105'),  
( '208', 'Nadiad', 'Kunj Raiyani', '9658745214', '107'),  
( '209', 'Jamnagar', 'Kalpesh Kaur', '9658745123', '108'),  
( '210', 'Bhavnagar', 'Priya Roy', '9564781243', '110');
```

5.2.3 Manager

```
INSERT INTO Manager (  
"M_id", "M_name", "M_city", "D_id_fk", "P_uid_fk")  
VALUES ('610', 'Yogesh Malhotra', 'Delhi', '509', '109'),  
( '609', 'Rajesh Parekh', 'MUmbai', '510', '110'),  
( '608', 'Abdul Shekh', 'Lahore', '507', '108'),  
( '607', 'Jin-Ping', 'Chaina', '506', '107'),  
( '606', 'M. Sui', 'Japan', '505', '106'),  
( '605', 'M. k. Yung', 'Landan', '504', '105'),  
( '604', 'Vipul Patel', 'Surendranagar', '503', '104'),  
( '603', 'Pravin Sojitra', 'Rajkot', '502', '103'),  
( '602', 'Balavant vaghani', 'Surat', '502', '102'),  
( '601', 'Prusthi Sorathiya', 'Ahmedabad', '501', '101');
```

5.2.4 Department

```
INSERT INTO Department (  
"D_id", "D_name", "E3_id_fk")  
VALUES ('510','Problem solver','210'), ('509','AI','208'), ('508','design','207'),  
('507','Coder','206'), ('506','Analysis','204'), ('505','Data','204'),  
('504','Data Base','203'), ('503','programmer','202'), ('502','software','201'),  
('501','Hardware','201');
```

5.2.5 Project

```
INSERT INTO Project (  
"P_id", "D_cost", "P_name", "E2_id_fk")  
VALUES ('501', '500000', 'Alpha', '201'), ('502', '100000', 'Beta', '202'),  
('503', '150000', 'Delta', '203'), ('504', '300000', 'Lemda', '204'),  
('505', '1000000', 'Scretch', '205'), ('506', '150000', 'Theta', '206'),  
('507', '100000', 'New', '207'), ('508', '180000', 'Sigma', '208'),  
('509', '190000', 'Solid', '209'), ('510', '200000', 'Tesla', '210');
```

5.2.6 Tasks

```
INSERT INTO Task (  
"T_id", "T_discription", start_date, end_date, "T_status", p_id_fk)  
VALUES ('401','Web Development','2024-01-01','2025-02-13','Completed'),  
('403','HArduware Development','202-06-24','2025-03-13','In Progress'),  
('404','Data analysis','2024-07-29','2025-06-24','In Progress'),  
('405','Web Development','2024-08-09','2025-02-08','Completed'),  
('406','Programmer','2024-09-01','2025-01-08','Completed'),  
('407','Data base Development','2024-03-13','2025-02-27','Completed'),  
('408','Hardware Development','2025-01-24','2025-04-15','In Progress'),
```

```
( '409','software Development','2024-07-21','2025-01-17','Completed'),
( '410','Programer','2024-05-10','2025-02-09','Completed'),
( '402','Software Development','2024-04-10','2025-01-24','Completed');
```

5.2.7 Leaves

```
INSERT INTO Leave (
    "L_id", "L_status", "L_date", "L_message", "E1_id_fk")
VALUES ( '301', 'Pending', '2025-02-27', 'For Wedding Purpose', '201'),
( '302', 'Approved', '2025-01-15', 'For Rest', '202'),
( '303', 'Approved', '2025-01-31', 'For Personal Reason', '202'),
( '304', 'Approved', '2025-02-08', 'For Wedding Purpose', '203'),
( '305', 'Pending', '2025-02-08', 'For Rest Purpose', '204'),
( '306', 'Pending', '2025-01-23', 'For Wedding Purpose', '205'),
( '307', 'Approved', '2025-02-20', 'For Personal Reason', '206'),
( '308', 'Approved', '2025-01-24', 'For Personal Reason', '207'),
( '309', 'Approved', '2025-02-06', 'For Wedding Purpose', '208'),
( '310', 'Pending', '2025-01-22', 'For Wedding Purpose', '201');
```

5.2.8 Tasks and Project combine

```
INSERT INTO PandTcombine (
    "P_T_id", "P_id_fk", "T_id_fk")
VALUES ( '701', '501', '401'), ( '702', '502', '403'), ( '703', '504', '402'),
( '704', '503', '405'), ( '705', '506', '404'), ( '706', '507', '406'), ( '707', '508', '407'),
( '708', '509', '410'), ( '709', '510', '409'), ( '710', '510', '408'), ( '711', '501', '401'),
( '712', '504', '403'), ( '713', '506', '402'), ( '714', '503', '405'), ( '715', '502', '404'),
( '716', '507', '407'), ( '717', '509', '406'), ( '718', '508', '408'), ( '719', '506', '410'),
( '720', '502', '409');
```

INSERTION OUTPUT:

5.2.1 Person

```
Employee_Manager_Management_System=# select * from "Person";
P_uid | P_email | P_city | P_name
-----+-----+-----+-----
108 | raj1@gmail.com | Surat | Raj Shah
109 | jay1@gmail.com | Rajkot | jay sakriya
110 | yug3@gmail.com | Surendranagar | Yug Sakriya
107 | jaypal4@gmail.com | Nadiad | Jaypal Soni
106 | Sandip5@gmail.com | Valsad | Sandip Sojitra
105 | Rahul3@gmail.com | Jamnagar | Rahul Sakriya
104 | Dev5@gmail.com | Navsari | Dev Ansari
103 | Deven1@gmail.com | Surat | Deven Gol
102 | om12@gmail.com | Ahmedabad | Om Jani
101 | Jatin7@gmail.com | Palitana | Jatin mandali
(10 rows)
```

5.2.2 Employee

```
Employee_Manager_Management_System=# select * from "Employee";
E_id | E_city | E_name | E_m_number | P_uid_fk
-----+-----+-----+-----+-----
210 | Bhavnagar | Priya Roy | 9564781243 | 110
209 | Jamnagar | Kalpesh Kaur | 9658745123 | 108
208 | Nadiad | Kunj Raiyani | 9658745214 | 107
207 | Dang | Ansh Thummar | 9658742145 | 105
206 | Bhuj | Vedant Patel | 9321456854 | 106
205 | Rajkot | Vinod Kambl | 9875463215 | 103
204 | vadodara | Suresh Raina | 9987563214 | 103
203 | Nadiad | Irfan Pathan | 9587456325 | 101
202 | Surat | Yusuf Pathan | 9745786165 | 102
201 | Ahmedabad | Akshat jani | 9874567813 | 101
(10 rows)
```

5.2.3 Manager

```
Employee_Manager_Management_System=# select * from "Manager";
M_id | M_name | M_city | D_id_fk | P_uid_fk
-----+-----+-----+-----+-----
610 | Yogesh Malhotra | Delhi | 509 | 109
609 | Rajesh Parekh | Mumbai | 510 | 110
608 | Abdul Shekh | Lahore | 507 | 108
607 | Jin-Ping | China | 506 | 107
606 | M. Sui | Japan | 505 | 106
605 | M. K. Yung | Landan | 504 | 105
604 | Vipul Patel | Surendranagar | 503 | 104
603 | Pravin Sojitra | Rajkot | 502 | 103
602 | Balavant vaghani | Surat | 502 | 102
601 | Prusthi Sorathiya | Ahmedabad | 501 | 101
(10 rows)
```

5.2.4 Department

```
Employee_Manager_Management_System=# select * from "Department";
 D_id |  D_name  | E3_id_fk
-----+-----+-----
 510 | Problem solver | 210
 509 | AI       | 208
 508 | design  | 207
 507 | Coder   | 206
 506 | Analysis | 204
 505 | Data    | 204
 504 | Data Base | 203
 503 | programmer | 202
 502 | software | 201
 501 | Hardware | 201
(10 rows)
```

5.2.5 Project

```
Employee_Manager_Management_System=# select * from "Project";
 P_id | P_cost | P_name | E2_id_fk
-----+-----+-----+-----
 510 | 200000 | Tesla | 210
 509 | 190000 | Solid | 209
 508 | 180000 | Sigma | 208
 507 | 100000 | New   | 207
 506 | 150000 | Theta | 206
 505 | 1000000 | Scretch | 205
 504 | 300000 | lemnda | 204
 503 | 150000 | Delta | 203
 502 | 100000 | Beta | 202
 501 | 500000 | Alpha | 201
(10 rows)
```

5.2.6 Tasks

```
Employee_Manager_Management_System=# select * from "Task";
 T_id | T_discription | start_date | end_date | T_status | p_id_fk
-----+-----+-----+-----+-----+-----
 401 | Web Development | 2024-01-01 | 2025-02-13 | Completed | 501
 402 | software Development | 2024-04-10 | 2025-01-24 | Completed | 501
 403 | Hardware Development | 2024-06-24 | 2025-03-13 | In Progress | 502
 404 | Data analysis | 2024-07-29 | 2025-06-24 | In Progress | 504
 405 | Web Development | 2024-08-09 | 2025-02-08 | Completed | 503
 406 | Programmer | 2024-09-01 | 2025-01-08 | Completed | 509
 407 | Data Base Development | 2024-03-13 | 2025-02-27 | completed | 510
 408 | Hardware development | 2025-01-24 | 2025-04-15 | In progress | 507
 409 | software Development | 2024-07-21 | 2025-01-17 | completed | 508
 410 | Programmer | 2024-05-10 | 2025-02-09 | Completed | 503
 411 | Test task | 2025-03-05 | 2025-03-10 | Pending | 510
(11 rows)
```

5.2.7 Leaves

```
Employee_Manager_Management_System=# select * from "Leave";
 l_id | l_status | l_date | l_message | El_id_fk
-----+-----+-----+-----+-----
 301 | Pending | 2025-02-27 | For Wedding Prupose | 201
 302 | Aproved | 2025-01-15 | For Rest | 202
 303 | Aproved | 2025-01-31 | For personal Reason | 202
 304 | Aproved | 2025-02-08 | For Wedding Purpose | 203
 305 | Pending | 2025-02-08 | For Rest Prupose | 204
 306 | Pending | 2025-01-23 | For Wedding Purpose | 205
 307 | Aproved | 2025-02-20 | For Personal Reason | 206
 308 | Aproved | 2025-01-24 | For Personal reason | 207
 309 | Aproved | 2025-02-06 | For Wedding Purpose | 208
 310 | Peding | 2025-01-22 | For Wedding Purpose | 201
(10 rows)
```

5.2.8 Project and Tasks Combine

```
Employee_Manager_Management_System=# select * from "PandTcombine";
 P_T_id | P_id_fk | T_id_fk
-----+-----+-----
 720 | 502 | 409
 719 | 506 | 410
 718 | 508 | 408
 717 | 509 | 406
 716 | 507 | 407
 715 | 502 | 404
 714 | 503 | 405
 713 | 506 | 402
 712 | 504 | 403
 711 | 501 | 401
 710 | 510 | 408
 709 | 510 | 409
 708 | 509 | 410
 707 | 508 | 407
 706 | 507 | 406
 705 | 506 | 404
 704 | 503 | 405
 703 | 504 | 402
 702 | 502 | 403
 701 | 501 | 401
(20 rows)
```

5.3 QUERIES USING BASIC DBMS CONSTRUCTS

JOIN & SUBQUERIES:

5.3.1 Display all employee names who lives in Nadiad

```
HINT: Perhaps you meant to reference the column "Employee.E_city".
Employee_Manager_Management_System=# select "E_name" from "Employee" where "E_city"='Nadiad';
 E_name
-----
Kunj Raiyani
Irfan Pathan
(2 rows)
```

5.3.2 Count total no of leaves taken by every employee

```
Employee_Manager_Management_System=# select "E1_id_fk", count(*) from "Leave" group by "E1_id_fk";
 E1_id_fk | count
-----|-----
      204 |      1
      203 |      1
      205 |      1
      201 |      2
      207 |      1
      208 |      1
      206 |      1
      202 |      2
(8 rows)
```

5.3.3 Display employee name along with their project and department

```
Employee_Manager_Management_System=# select "P_name" as project_name,"E_name" as employee_name,"D_name" as department_name from "Project" join "Employee" on "E2_id_fk"="E_id" join "Department" on "E_id"="E3_id_fk" order by "P_name","E_name";
 project_name | employee_name | department_name
-----|-----|-----
Alpha        | Akshat jani   | Hardware
Beta         | Yusuf Pathan | programmer
Delta        | Irfan Pathan  | Data Base
Lenda        | Suresh Raina  | Data
New          | Ansh Thummar  | design
Scretch      | Vinod Nambl   | Analysis
Sigma        | Kunj Raiyani  | AI
Solid        | Malpesh Maur  | software
Tesla        | Priya Roy     | Problem solver
Theta        | Vedant Patel  | Coder
(10 rows)
```

5.3.4 Display details of all manager

```
Employee_Manager_Management_System=# select * from "Manager";
 M_id |  M_name  |  M_city  | D_id_fk | P_uid_fk
-----+-----+-----+-----+-----
 610 | Yogesh Malhotra | Delhi | 509 | 109
 609 | Rajesh Parekh | Mumbai | 510 | 110
 608 | Abdul Shekh | Lahore | 507 | 108
 607 | Jin-Ping | Chaina | 506 | 107
 606 | M. Sui | Japan | 505 | 106
 604 | Vipul Patel | Surendranagar | 503 | 104
 602 | Balavant vaghani | Surat | 502 | 102
 601 | Prusthi Sorathiya | Ahmedabad | 501 | 101
 605 | M. K. Yung | London | 504 | 105
 603 | Pravin Sojitra | Rajkot | 508 | 103
(10 rows)
```

5.3.5 Display five manager details in descending order

```
Employee_Manager_Management_System=# select * from "Manager" order by "M_id" desc limit 5;
 M_id |  M_name  |  M_city  | D_id_fk | P_uid_fk
-----+-----+-----+-----+-----
 610 | Yogesh Malhotra | Delhi | 509 | 109
 609 | Rajesh Parekh | Mumbai | 510 | 110
 608 | Abdul Shekh | Lahore | 507 | 108
 607 | Jin-Ping | Chaina | 506 | 107
 606 | M. Sui | Japan | 505 | 106
(5 rows)
```

5.3.6 Display employee id, name and project name whose project id is 302

```
(5 rows)

Employee_Manager_Management_System=# select "E_id", "E_name", "P_name" from "Employee" join "Project" on "E_id" = "E2_id_fk" where "P_id"=302;
 E_id | E_name | P_name
-----+-----+-----
(0 rows)
```

5.3.7 Display all employee ids

```
Employee_Manager_Management_System=# select "E_id" from "Employee";
E_id
-----
 209
 208
 207
 206
 205
 204
 203
 202
 201
 210
(10 rows)
```

5.3.8 Display project details with project cost in greater than 100000

```
Employee_Manager_Management_System=# select * from "Project" where "P_cost">'100000';
P_id | P_cost | P_name | E2_id_fk
-----+-----+-----+-----
 510 | 200000 | Tesla | 210
 509 | 190000 | Solid | 209
 508 | 180000 | Sigma | 208
 506 | 150000 | Theta | 206
 505 | 1000000 | Scretch | 205
 504 | 300000 | lemدا | 204
 503 | 150000 | Delta | 203
 501 | 500000 | Alpha | 201
(8 rows)
```

5.3.9 Display all task details which was completed before 13th march

```
Employee_Manager_Management_System=# select * from "Task" where "start_date"<'2025-03-13';
T_id | T_discription | start_date | end_date | T_status | p_id_fk
-----+-----+-----+-----+-----+-----
 401 | Web Development | 2024-01-01 | 2025-02-13 | Completed | 501
 402 | software Development | 2024-04-10 | 2025-01-24 | Completed | 501
 403 | Hardware Development | 2024-06-24 | 2025-03-13 | In Progress | 502
 404 | Data analysis | 2024-07-29 | 2025-06-24 | In Progress | 504
 405 | Web Development | 2024-08-09 | 2025-02-08 | Completed | 503
 406 | Programmer | 2024-09-01 | 2025-01-08 | Completed | 509
 407 | Data Base Development | 2024-03-13 | 2025-02-27 | completed | 510
 408 | Hardware development | 2025-01-24 | 2025-04-15 | In progress | 507
 409 | software Development | 2024-07-21 | 2025-01-17 | completed | 508
 410 | Programmer | 2024-05-10 | 2025-02-09 | Completed | 503
(10 rows)
```

5.3.10 Display all tasks details where status in completed

```
Employee_Manager_Management_System=# select "T_id","T_discription","start_date","end_date" from "Task" where "T_status"='Completed';
```

T_id	T_discription	start_date	end_date
401	Web Development	2024-01-01	2025-02-13
402	software Development	2024-04-10	2025-01-24
405	Web Development	2024-08-09	2025-02-08
406	Programmer	2024-09-01	2025-01-08
410	Programmer	2024-05-10	2025-02-09
407	Data Base Development	2024-03-13	2025-02-27
409	software Development	2024-07-21	2025-01-17

(7 rows)

5.3.11 Display Project name, tasks with their starting & ending date

```
Employee_Manager_Management_System=# select "P_name","T_discription","start_date","end_date" from "Project" join "Task" on "P_id"="p_id_fk";
```

P_name	T_discription	start_date	end_date
Alpha	Web Development	2024-01-01	2025-02-13
Alpha	software Development	2024-04-10	2025-01-24
Beta	Hardware Development	2024-06-24	2025-03-13
lemda	Data analysis	2024-07-29	2025-06-24
Delta	Web Development	2024-08-09	2025-02-08
Solid	Programmer	2024-09-01	2025-01-08
New	Hardware development	2025-01-24	2025-04-15
Delta	Programmer	2024-05-10	2025-02-09
Tesla	Test task	2025-03-05	2025-03-10
Tesla	Data Base Development	2024-03-13	2025-02-27
Sigma	software Development	2024-07-21	2025-01-17

(11 rows)

5.3.12 Find name of the employee who have taken max leave

```
Employee_Manager_Management_System=# with LeaveCounts as (select "E_name",count("E1_id_fk") as leave_count from "Leave" join "Employee" on "E1_id_fk"="E_id" group by "E_name")select "E_name",leave_count from LeaveCounts where leave_count=(select max(leave_count) from LeaveCounts);
```

E_name	leave_count
Akshat jani	2
Yusuf Pathan	2

(2 rows)

5.3.13 Find name of the employee who is working on project task starting from 1st January

```
Employee_Manager_Management_System=# select distinct "E_name" from "Employee" where "E_id" in (select "E2_id_fk" from "Project" where "P_id" in (select "p_id_fk" from "Task" where "start_date"='2024-01-01'));
      E_name
-----
Akshat jani
(1 row)
```

5.3.14 count the total no of employees in each department

```
Employee_Manager_Management_System=# select "D_name",count("E_id") as total_employees from "Employee" join "Department"
on "E3_id_fk"="E_id" group by "D_name";
      D_name      | total_employees
-----+-----
Data               | 1
software           | 1
Data Base          | 1
AI                 | 1
Problem solver     | 1
Hardware           | 1
design              | 1
Coder              | 1
programmer         | 1
Analysis           | 1
(10 rows)
```

5.3.15 Count the total no of employees from Employee table

```
Employee_Manager_Management_System=# select count(*) as total_employees from "Employee";
      total_employees
-----
10
(1 row)
```

5.4 PL/SQL:

5.4.1 View

```
Employee_Manager_Management_System=# create view E_name_city as select "E_id","E_city" from "Employee";
CREATE VIEW
Employee_Manager_Management_System=# select * from E_name_city;
 E_id | E_city
-----+-----
 209 | Jamnagar
 208 | Nadiad
 207 | Dang
 206 | Bhuj
 205 | Rajkot
 204 | vadodara
 203 | Nadiad
 202 | Surat
 201 | Ahmedabad
 210 | Bhavnagar
(10 rows)
```

5.4.2 Cursor

```
Employee_Manager_Management_System=# begin;
BEGIN
Employee_Manager_Management_System=# declare my_E_id cursor for
Employee_Manager_Management_System=# select "E_id" from "Employee" where "E_id">209;
DECLARE CURSOR
Employee_Manager_Management_System=# fetch next from my_E_id;
 E_id
-----
 210
(1 row)

Employee_Manager_Management_System=# close my_E_id;
CLOSE CURSOR
Employee_Manager_Management_System=#
```

5.4.3 Trigger

```
Employee_Manager_Management_System=# create or replace function check_employee_project() returns trigger as $$
Employee_Manager_Management_System$# declare employee_project_exists int;
Employee_Manager_Management_System$# begin
Employee_Manager_Management_System$# select count(*) into employee_project_exists from "Project" where "P_id"=new."p_id_fk";
Employee_Manager_Management_System$# if employee_project_exists=0 then
Employee_Manager_Management_System$# raise exception 'Employee must be assigned to a project before getting a task';
Employee_Manager_Management_System$# end if;
Employee_Manager_Management_System$# return new;
Employee_Manager_Management_System$# end;
Employee_Manager_Management_System$# $$ language plpgsql;
CREATE FUNCTION
Employee_Manager_Management_System=# insert into "Task"("T_id","T_description","start_date","end_date","T_status","p_id_fk") values('411','Test task','2025-03-05','2025-03-10','Pending','511');
ERROR: Employee must be assigned to a project before getting a task
CONTEXT: PL/pgSQL function check_employee_project() line 6 at RAISE
Employee_Manager_Management_System=#
```

5.4.4 Function

```
Employee_Manager_Management_System=# create function projectcostbetween(costfrom int, costto int)
Employee_Manager_Management_System=# returns int
Employee_Manager_Management_System=# language plpgsql
Employee_Manager_Management_System=# as
Employee_Manager_Management_System=# $$
Employee_Manager_Management_System$# declare
Employee_Manager_Management_System$# numberofproject integer;
Employee_Manager_Management_System$# begin
Employee_Manager_Management_System$# select count(*)
Employee_Manager_Management_System$# into numberofproject
Employee_Manager_Management_System$# from "Project"
Employee_Manager_Management_System$# where "P_cost" between costfrom and costto;
Employee_Manager_Management_System$# return numberofproject;
Employee_Manager_Management_System$# end;
Employee_Manager_Management_System$# $$;
CREATE FUNCTION
Employee_Manager_Management_System=# select projectcostbetween(100000,300000);
projectcostbetween
-----
      8
(1 row)
```